



ELA: Recursive Linguistic Elements & the Context–Pattern Duality

An AI-Assisted Framework for Authentic English Comprehension — Where does language end and pattern begin?

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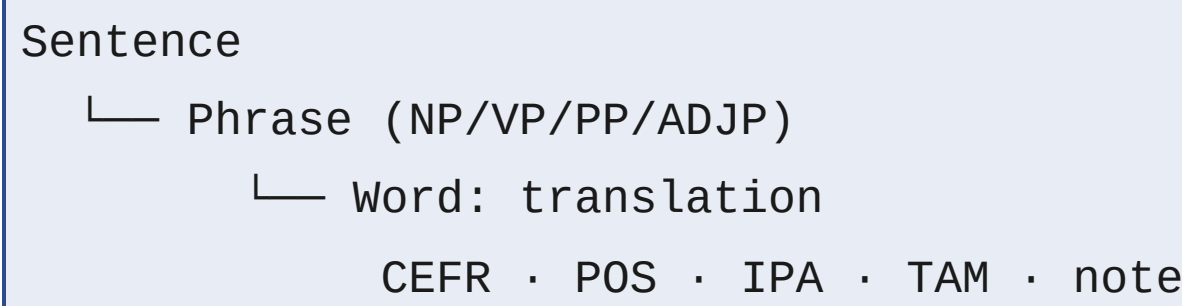


Problem & Motivation

Platforms simplify authentic input. We annotate it — **structure intact**.

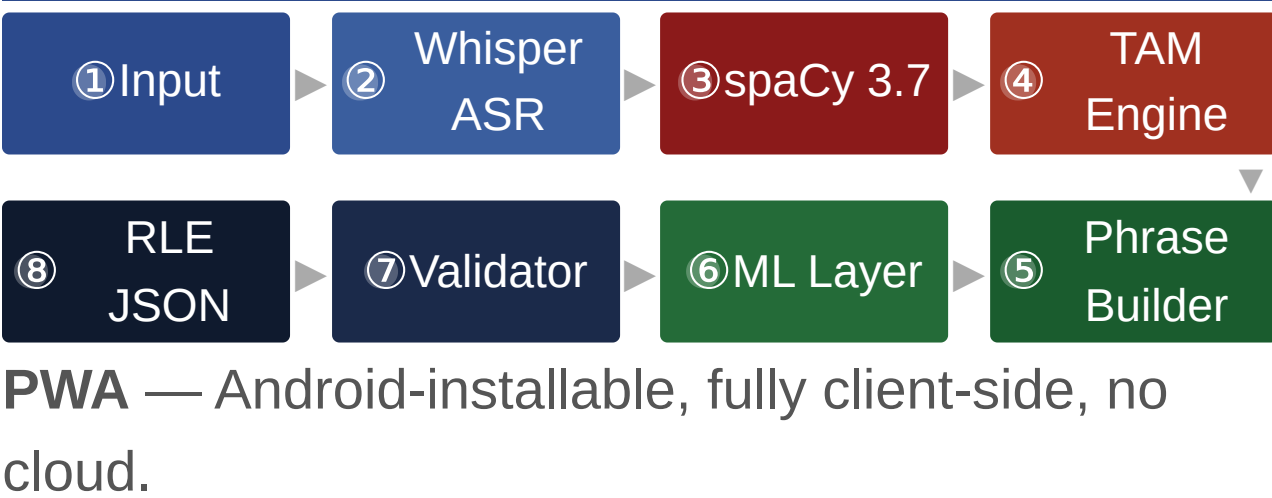
- Can NLP + AI explain English without simplifying it?
- What recursive schema stays human-interpretable?
- Does bilingual annotation scaffold comprehension?

The RLE Data Model



Language-agnostic · EN → RU via M2M100 (offline).

Analysis Pipeline



Research Journey: What We Tried

CEFR CLASSIFICATION		NOTE GENERATION	
DeBERTa underperformed	BERT same	GPT-4o cost/cloud	T5 raw no-generalise
XGBoost ✓ 78.3%		T5+Templates ✓ ROUGE-L 0.81	
ASR		TRANSLATION	
Whisper ✓ WER 8.2%		GPT-4o opt-in	M2M100 ✓ offline

Context–Pattern Interaction

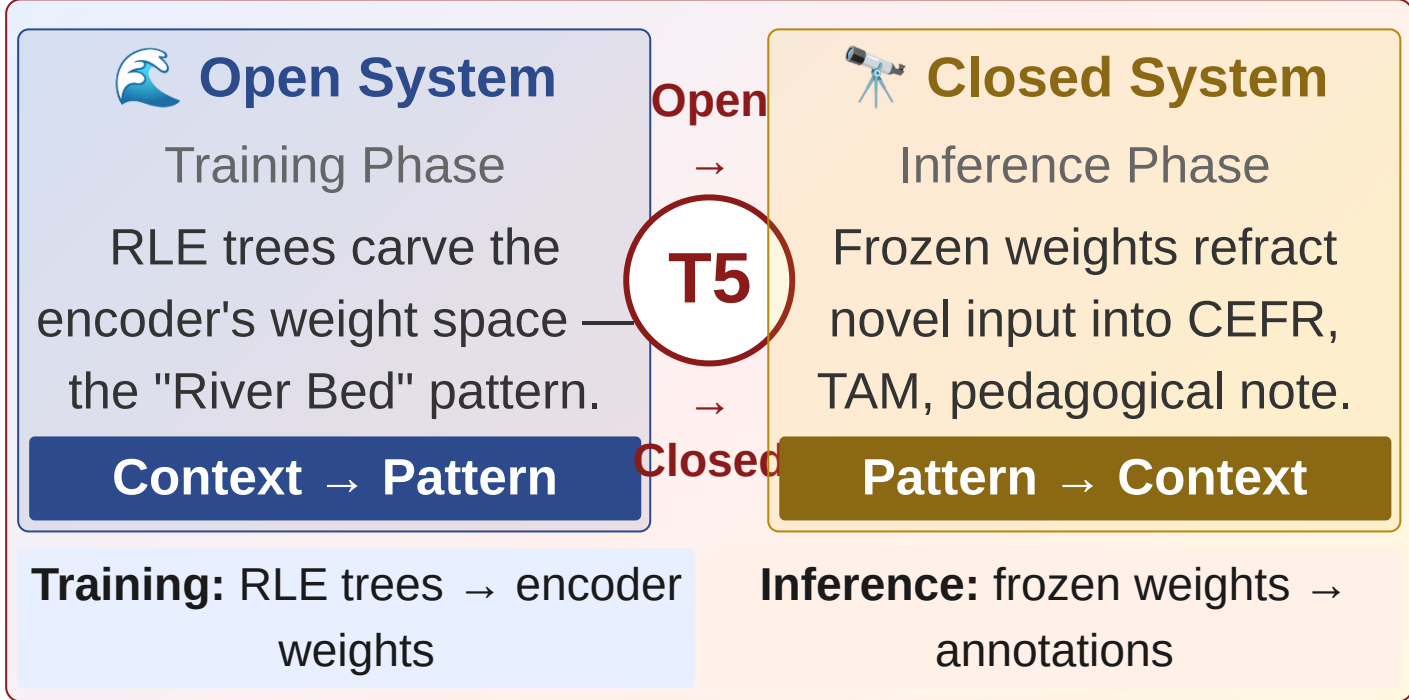
"In open systems, context forms patterns.
In closed systems, patterns form context."

— Discovered empirically during T5 fine-tuning

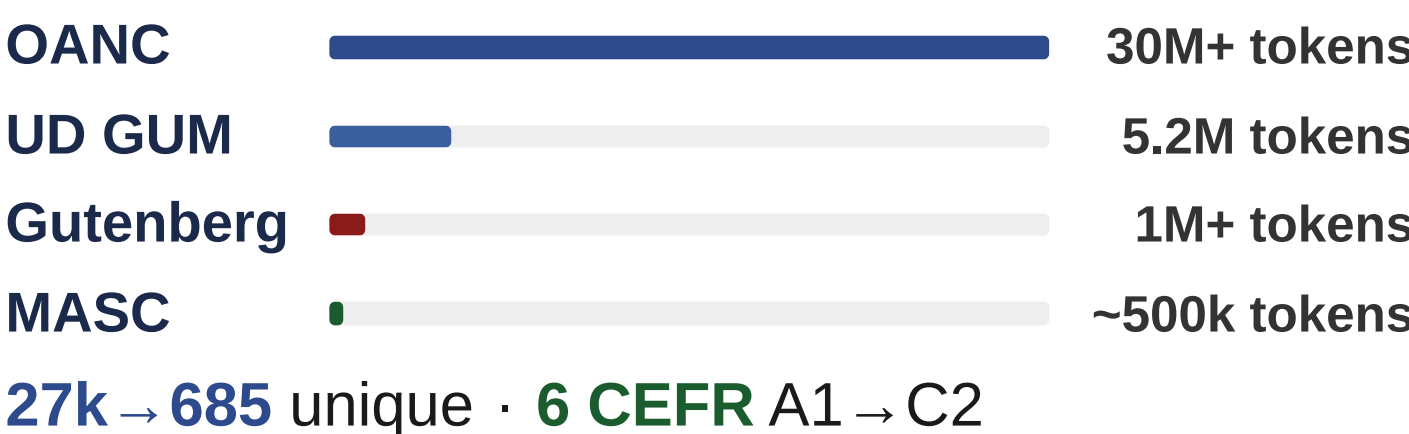
T5 on raw notation **failed**. Structural templates — **lexically invariant** — were the breakthrough:

✗ "gave him" → [VP gave][NP him]
✓ [V-PAST][IOBJ] → [TAM:past-simple]
[VOICE:active]

The Duality Loop Diagram



Training Data & Corpus Scale



Performance Metrics

78.3% CEFR Accuracy XGBoost ensemble	0.81 ROUGE-L T5 fine-tuned	8.2% ASR WER Whisper base
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Annotation Backend Comparison

Approach	Quality	Cost	Offline
GPT-4o	High	\$\$\$	✗
Rules only	Medium	Free	✓
T5 fine-tuned	High	Free	✓
XGBoost CEFR	78.3%	Free	✓

System Validation

- O1:** spaCy + TAM → schema-valid RLE ✓
- O2:** CEFR 78.3%; T5 ROUGE-L 0.81 ✓
- O3:** Visualizer + JSON/CSV export ✓
- O4:** Android PWA, fully client-side ✓

Conclusions

- AI annotation makes authentic English **accessible without simplification**.
- The **Context ↔ Pattern duality** generalises to any open → closed system; T5 placeholders are **lexically invariant**.

Future Directions

- Tauri 2 Desktop App** — native .deb/Windows with bundled ML models.
- Translation Quality Gates** — ChrF scorer, M2M100 review queue.
- M2M100 LoRA Fine-Tuning** — domain-specific EN → RU refinement.
- Real-time Streaming ASR** — Whisper streaming for live practice.

References

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Acknowledgements

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Live Demo

www.el-a.uk

Try it in your browser